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Semiconductor device and control circuit thereof

Abstract

According to one embodiment, a control circuit includes a first bias terminal, a second bias terminal, an input-side terminal, a diode, a capacitor, a first transistor, a second transistor, and an output-side terminal. The first transistor includes a first control terminal configured to turn on and off electrical conduction between a third terminal and a fourth terminal. The second transistor includes a second control terminal configured to turn on and off electrical conduction between a fifth terminal and a sixth terminal. A control signal based on a signal input to the input-side terminal is input to the first control terminal and the second control terminal, and the first transistor and the second transistor are alternately turned on and alternately turned off.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2023-045905, filed on Mar. 22, 2023; the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments relate to a semiconductor device and a control circuit thereof.

BACKGROUND

(3) A semiconductor device such as a MOS transistor is required to reduce on-resistance (conduction loss) and switching loss.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic view showing a semiconductor device according to a first embodiment.
- (2) FIG. 2 is a circuit diagram showing the semiconductor device according to the first embodiment.
- (3) FIG. 3 is a circuit diagram showing an operation of the semiconductor device in an off-state according to the first embodiment.
- (4) FIG. 4 is a circuit diagram showing an operation of the semiconductor device in an on-state according to the first embodiment.
- (5) FIG. 5 is a time chart showing a control method for the semiconductor device according to the first embodiment.
- (6) FIG. 6 is a time chart showing a control waveform of the semiconductor device according to the first embodiment.
- (7) FIG. 7 is a circuit diagram showing the control circuit according to the first embodiment.
- (8) FIG. 8A is a schematic cross-sectional view showing a control method for the semiconductor device according to a comparative example.
- (9) FIG. 8B is a schematic cross-sectional view showing a control method for the semiconductor device according to a comparative example.
- (10) FIG. 9A is a graph showing switching characteristics of the semiconductor device according to the first embodiment and the comparative example.
- (11) FIG. 9B is a graph showing switching characteristics of the semiconductor device according to the first embodiment and the comparative example.
- (12) FIG. 9C is a graph showing switching characteristics of the semiconductor device according to the first embodiment and the comparative example.
- (13) FIG. 10A is a graph showing another switching characteristic of the semiconductor device according to the first embodiment and the comparative example.
- (14) FIG. 10B is a graph showing another switching characteristic of the semiconductor device according to the first embodiment and the comparative example.
- (15) FIG. 10C is a graph showing another switching characteristic of the semiconductor device according to the first embodiment and the comparative example.
- (16) FIG. 11 is a circuit diagram showing a semiconductor device according to a variation of the first embodiment.
- (17) FIG. 12 is a circuit diagram showing a control circuit according to another variation of the first embodiment.
- (18) FIG. 13 is a time chart showing a control method for the semiconductor device according to another variation of the first embodiment.
- (19) FIG. 14 is a schematic cross-sectional view showing a semiconductor device according to a second embodiment.
- (20) FIG. 15A is a schematic cross-sectional view showing electrode connections in the semiconductor device according to the second embodiment.
- (21) FIG. 15B is a schematic cross-sectional view showing electrode connections in the semiconductor device according to the second embodiment.
- (22) FIG. 16 is a schematic cross-sectional view showing a semiconductor device according to a variation of the second embodiment.
- (23) FIG. 17 is a schematic view showing a semiconductor device according to a third

embodiment.

(24) FIG. **18** is a schematic view showing a semiconductor device according to a fourth embodiment.

DETAILED DESCRIPTION

(25) In general, according to one embodiment, a control circuit includes a first bias terminal, a second bias terminal separated from the first bias terminal, an input-side terminal separated from the first bias terminal and the second bias terminal, a diode having an anode connected to the first bias terminal, a capacitor including a first terminal connected to a cathode of the diode and a second terminal electrically connected to the second bias terminal, a first transistor including a third terminal connected to the first terminal of the capacitor, a fourth terminal, and a first control terminal configured to turn on and off electrical conduction between the third terminal and the fourth terminal, a second transistor including a fifth terminal connected to the fourth terminal of the first transistor, a sixth terminal connected to the second bias terminal, and a second control terminal configured to turn on and off electrical conduction between the fifth terminal and the sixth terminal, and an output-side terminal separated from the first bias terminal, the second bias terminal, and the input-side terminal and connected to the fourth terminal of the first transistor and the fifth terminal of the second transistor. a control signal based on a signal input to the input-side terminal being input to the first control terminal and the second control terminal, and the first transistor and the second transistor being alternately turned on and alternately turned off.

(26) Hereinafter, embodiments will be described with reference to the drawings. The detailed description of the same portion in the drawings attached with the same reference sign will be omitted as appropriate, and a different portion will be described. The drawings are schematic or conceptual. A relationship between a thickness and a width of each portion, a ratio of sizes between portions, and the like are not necessarily the same as the actual ones. Even if same portions are shown, dimensions and ratios may be shown differently from each other in the drawings.

(27) Next, an arrangement and a configuration of each part will be described using an X axis, a Y axis, and a Z axis shown in drawings. The X axis, the Y axis, and the Z axis are orthogonal to one another and represent an X-direction, a Y-direction, and a Z-direction, respectively. The Z-direction may be described as an upper side, and an opposite direction of the Z-direction may be described as a lower side.

First Embodiment

(28) FIG. **1** is a schematic view showing a semiconductor device **1** according to a first embodiment. The semiconductor device **1** includes, for example, a switching element SD and a control circuit CC. The switching element SD is, for example, a power MOS transistor having a trench gate structure.

(29) As shown in FIG. **1**, the switching element SD includes, for example, a semiconductor portion **10**, a first electrode **20**, a second electrode **30**, control electrodes **40**, and third electrodes **50**. The semiconductor portion **10** is located between the first electrode **20** and the second electrode **30**. For example, the second electrode **30** faces the first electrode **20** with the semiconductor portion **10** interposed therebetween.

(30) For example, the first electrode **20** is provided on a back surface of the semiconductor portion **10**. The first electrode **20** is, for example, a drain electrode. For example, the second electrode **30** is provided on an opposite-side surface from the back surface of the semiconductor portion **10**. The second electrode **30** is, for example, a source electrode.

(31) The semiconductor portion **10** includes a first conductivity type first semiconductor layer **11**, a second conductivity type second semiconductor layer **13**, a first conductivity type third semiconductor layer **15**, a second conductivity type fourth semiconductor layer **17**, and a first conductivity type fifth semiconductor layer **19**. Hereinafter, the first conductivity type will be described as an n-type, and the second conductivity type will be described as a p-type. However, the embodiment is not limited thereto.

(32) The first semiconductor layer **11** extends between the first electrode **20** and the second electrode **30**. The first semiconductor layer **11** is, for example, an n-type drift layer. The second semiconductor layer **13** is provided between the first semiconductor layer **11** and the second electrode **30**. The second semiconductor layer **13** is, for example, a p-type body layer.

(33) The third semiconductor layer **15** is provided between the second semiconductor layer **13** and the second electrode **30**. The third semiconductor layer **15** is, for example, an n-type source layer. For example, the third semiconductor layer **15** is in contact with and electrically connected to the second electrode **30**.

(34) The fourth semiconductor layer **17** is provided between the second semiconductor layer **13** and the second electrode **30** and is at least partially located in the second semiconductor layer **13**. The fourth semiconductor layer **17** is, for example, a p-type contact layer. The fourth semiconductor layer **17** contains a second conductivity type impurity having a concentration higher than a concentration of a second conductivity type impurity in the second semiconductor layer **13**. The second electrode **30** includes, for example, a contact portion **30c** that is in contact with and electrically connected to the fourth semiconductor layer **17**. The second electrode **30** is electrically connected to the second semiconductor layer **13** via the fourth semiconductor layer **17**.

(35) The fifth semiconductor layer **19** is provided between the first semiconductor layer **11** and the first electrode **20**. The fifth semiconductor layer **19** is, for example, an n-type buffer layer. The fifth semiconductor layer **19** contains an n-type impurity having a concentration higher than a concentration of an n-type impurity in the first semiconductor layer **11**. For example, the first electrode **20** is in contact with and electrically connected to the fifth semiconductor layer **19**.

(36) The semiconductor portion **10** has a trench TG provided on a surface side thereof. The control electrodes **40** and the third electrodes **50** are provided inside the trench TG. The control electrodes **40** are located between the second electrode **30** and the third electrodes **50**. The third electrodes **50** are located between the first electrode **20** and the control electrodes **40**.

(37) The control electrode **40** is, for example, a gate electrode. The control electrodes **40** are provided at positions facing the second semiconductor layer **13** via first insulating films **43**. The first insulating film **43** is, for example, a gate insulating film. The control electrodes **40** face the first semiconductor layer **11** via the first insulating films **43**. The control electrodes **40** also face the third semiconductor layer **15** via the first insulating films **43**. In other words, the first insulating film **43** electrically insulates the control electrode **40** from the semiconductor portion **10**.

(38) Further, second insulating films **45** are provided between the second electrode **30** and the control electrodes **40**. The second insulating film **45** is, for example, an interlayer insulating film. The second insulating film **45** electrically insulates the control electrode **40** from the second electrode **30**.

(39) The third electrode **50** is, for example, a field plate. The third electrodes **50** are located in the first semiconductor layer **11**. The third electrodes **50** face the first semiconductor layer **11** via third insulating films **53**. The third insulating film **53** is, for example, a field plate insulating film (an FP insulating film). The third insulating films **53** electrically insulate the third electrodes **50** from the first semiconductor layer **11**. Fourth insulating films **55** are provided between the control electrodes **40** and the third electrodes **50**. The fourth insulating films **55** electrically insulate the control electrodes **40** from the third electrodes **50**.

(40) As shown in FIG. 1, the semiconductor device **1** further includes a first terminal DT, a second terminal ST, and a control terminal GT. The first terminal DT, the second terminal ST, and the control terminal GT are connected to the switching element SD and the control circuit CC.

(41) The first terminal DT is, for example, a drain terminal. The first terminal DT is connected to the first electrode **20** of the switching element SD. The second terminal ST is, for example, a source terminal. The second terminal ST is connected to the second electrode **30** of the switching element SD. The control terminal GT is, for example, a gate terminal. The control terminal GT is electrically connected to, for example, the control electrode **40** of the switching element SD. The

control terminal GT is electrically connected to the control electrode **40** via, for example, a gate resistor Rg. The gate resistor Rg is, for example, an internal resistor of the control electrode **40**.

(42) For example, the control circuit CC is electrically connected to the first terminal DT, the second terminal ST, the control terminal GT, and the third electrode **50** of the switching element SD. The control circuit CC applies a field plate voltage VFP between the second electrode **30** and the third electrode **50**.

(43) The control circuit CC includes, for example, a first transistor Tr1, a second transistor Tr2, a third transistor Tr3, a first diode D1, a second diode D2, and a capacitor CB. The first transistor Tr1 is, for example, a PMOS transistor. The second transistor Tr2 and the third transistor Tr3 are, for example, NMOS transistors.

(44) The first diode D1 and the capacitor CB are connected in series between the first terminal DT and the second terminal ST. An anode of the first diode D1 is connected to the first terminal DT, and a cathode of the first diode D1 is connected to one terminal CBD of the capacitor CB. The other terminal CBS of the capacitor CB is connected to the second terminal ST.

(45) The first transistor Tr1 and the second transistor Tr2 are connected in series, and a drain of the first transistor Tr1 is connected to a drain of the second transistor Tr2. A source of the first transistor Tr1 is connected to the cathode of the first diode D1 and the terminal CBD of the capacitor CB. A source of the second transistor Tr2 is connected to the second terminal ST.

(46) The second diode D2 and the third transistor Tr3 are connected in series between the first terminal DT and the second terminal ST. An anode of the second diode D2 is connected to the first terminal DT, and the cathode of the second diode D1 is connected to a drain of the third transistor Tr3. A source of the third transistor is connected to the second terminal ST. A gate of the first transistor Tr1 and a gate of the second transistor Tr2 are connected to a cathode of the second diode D2 and the drain of the third transistor Tr3.

(47) A gate of the third transistor Tr3 is connected to the control terminal GT. The drain of the first transistor Tr1 and the drain of the second transistor Tr2 are electrically connected to the third electrode **50** of the switching element SD.

(48) FIG. 2 is a circuit diagram showing the semiconductor device **1** according to the first embodiment. As shown in FIG. 2, the switching element SD and the control circuit CC are connected to the first terminal DT and the second terminal ST, and are biased by a voltage Vds (see FIG. 5) applied between the first terminal DT and the second terminal ST.

(49) A drain D of the switching element SD is connected to the first terminal DT, and a source S of the switching element SD is connected to the second terminal ST. Further, a gate G of the switching element SD is connected to the control terminal GT.

(50) The switching element SD is controlled to be turned on and off by a control signal Vg (see FIG. 5) input to the gate G. The control signal Vg is input to the control terminal GT. The control terminal GT is electrically connected to the gate G of the switching element SD. The control signal Vg is supplied, for example, between the control terminal GT and the source S of the switching element SD.

(51) The control signal Vg is also input to the gate of the third transistor Tr3 via the control terminal GT. The third transistor Tr3 is controlled to be turned on and off based on the control signal Vg. Further, the first transistor Tr1 and the second transistor Tr2 are controlled to be turned on and off based on a potential on a drain side of the third transistor Tr3.

(52) For example, when the third transistor Tr3 is in an off-state and the drain side of the third transistor Tr3 is at a high potential “High”, the first transistor Tr1 is in an off-state and the second transistor Tr2 is in an on-state. When the third transistor Tr3 is in an on-state, the drain side of the third transistor Tr3 becomes a low potential “Low”, the first transistor Tr1 transitions from the off-state to the on-state, and the second transistor Tr2 transitions from the on-state to the off-state.

(53) FIG. 3 is a circuit diagram showing an operation of the semiconductor device **1** in an off-state according to the first embodiment. At this time, the control signal Vg (see FIG. 5) input to the

control terminal GT is “Low”. The control signal V_g is applied to the gate G of the switching element SD via an external gate resistor R_{ge} , and the switching element SD is in the off-state. A voltage V_{dd} is applied between the first terminal DT and the second terminal ST via a load resistor R_L . A drain voltage V_{d-off} (see FIG. 5) is applied between the first terminal DT and the second terminal ST.

(54) When the control signal V_g is “Low”, the third transistor Tr_3 of the control circuit CC is in the off-state (see FIG. 2), and the potential on the drain side of the third transistor Tr_3 is “High”. At this time, the first transistor Tr_1 is in the off-state, and the second transistor Tr_2 is in the on-state. The capacitor CB is charged by the drain voltage V_{d-off} . A field plate FP (the third electrode 50) of the switching element SD has a potential same as that of the second terminal ST via the second transistor Tr_2 in the on-state. That is, a potential difference V_{FP} between the field plate FP and the source S of the switching element SD is 0 V (see FIG. 5).

(55) FIG. 4 is a circuit diagram showing an operation of the semiconductor device 1 in an on-state according to the first embodiment. At this time, the control signal V_g (see FIG. 5) input to the control terminal GT is “High”. The gate G of the switching element SD becomes “High”, and the switching element SD is in the on-state. A drain current I_d flows between the first terminal DT and the second terminal ST, and the voltage between the first terminal DT and the second terminal ST decreases to a level of an on-voltage V_{d-on} (see FIG. 5).

(56) When the control signal V_g is “High”, the third transistor Tr_3 (see FIG. 2) of the control circuit CC is in the on-state, and the potential on the drain side of the third transistor Tr_3 is “Low”. At this time, the first transistor Tr_1 is in the on-state, and the second transistor Tr_2 is in the off-state. Therefore, electrical connection between the field plate FP of the switching element SD and the second terminal ST is disconnected, and the terminal CBD of the capacitor CB and the field plate FP are electrically connected via the first transistor Tr_1 in the on-state. Accordingly, charges in the capacitor CB move to the field plate FP, and the field plate FP has a potential same as that of the terminal CBD of the capacitor CB. That is, a voltage V_{c-on} between both terminals of the capacitor CB is applied as V_{fp-on} between the field plate FP and the source S of the switching element SD (see FIG. 5).

(57) FIG. 5 is a time chart showing a control method for the semiconductor device 1 according to the first embodiment. FIG. 5 shows a gate input signal V_{g-in} , a gate output signal V_{g-out} , a control signal V_{g1} of the first transistor Tr_1 , a control signal V_{g2} of the second transistor Tr_2 , a source-drain voltage V_{ds} , an inter-terminal voltage V_{CB} of the capacitor CB, and the field plate voltage V_{FP} .

(58) As shown in FIG. 5, the gate input signal V_{g-in} input to the control terminal GT increases, for example, from Low to High at a time T_1 . The gate output signal V_{g-out} output from the control circuit CC is equal to the gate input signal V_{g-in} , and is applied between the gate G and the source S of the switching element SD. Accordingly, the switching element SD transitions from the off-state to the on-state (hereinafter, turned on).

(59) The gate input signal V_{g-in} is also input to the gate of the third transistor Tr_3 , and the third transistor Tr_3 is turned on at the time T_1 . Therefore, the potential on the drain side of the third transistor Tr_3 changes from High to Low, and the control signal V_{g1} of the first transistor Tr_1 and the control signal V_{g2} of the second transistor Tr_2 also change from High to Low. Accordingly, the first transistor Tr_1 is turned on, and the second transistor Tr_2 transitions from the on-state to the off-state (hereinafter, turned off).

(60) The source-drain voltage V_{ds} decreases from an off-voltage V_{d-off} to the on-voltage V_{d-on} at the time T_1 . Since the on-voltage V_{d-on} is lower than a voltage V_{c-off} of the capacitor CB when the switching element SD is in the off-state, charging of the capacitor CB is stopped at the time T_1 .

(61) The capacitor CB is electrically connected to the field plate FP of the switching element SD via the first transistor Tr_1 in the on-state. On the other hand, the electrical connection between the field plate FP and the source S is disconnected by turning off the second transistor Tr_2 . Therefore,

the charges of the capacitor CB move to the field plate FP via the first transistor Tr1. Accordingly, the inter-terminal voltage of the capacitor CB decreases to the voltage V_{c-on} when the switching element SD is in the on-state.

(62) A parasitic capacitance of the field plate FP in the switching element SD is charged by the charges moving from the capacitor CB until the potential difference between the field plate FP and the source S is equal to the inter-terminal voltage VCB of the capacitor CB. Accordingly, the field plate voltage VFP increases from 0 V to V_{fp-on} at the time T1. V_{fp-on} is equal to the voltage V_{c-on} of the capacitor CB.

(63) Further, the gate input signal V_{g-in} changes from High to Low at a time T2. The gate output signal V_{g-out} also changes from High to Low, and the switching element SD is turned off. The third transistor Tr3 is also turned off. Therefore, the potential on the drain side of the third transistor Tr3 changes from Low to High, and the control signal V_{g1} of the first transistor Tr1 and the control signal V_{g2} of the second transistor Tr2 also change from Low to High. Accordingly, the first transistor Tr1 is turned off and the second transistor Tr2 is turned on.

(64) At the time T2, the source-drain voltage V_{ds} increases from the on-voltage V_{d-on} to the off-voltage V_{d-off} , and charging of the capacitor CB is started. The inter-terminal voltage VCB of the capacitor CB increases from V_{c-on} to V_{c-off} . The electrical connection between the capacitor CB and the field plate FP of the switching element SD is cut off by the first transistor Tr1 which is turned off, and the field plate FP and the source S are electrically connected via the second transistor Tr2 which is turned on. Therefore, the field plate voltage VFP between the field plate FP and the source S is 0 V.

(65) Subsequently, the gate input signal V_{g-in} changes from Low to High at a time T3, and the switching element SD and the third transistor Tr3 are turned on. Accordingly, the first transistor Tr1 is turned on, and the second transistor Tr2 is turned off. As a result, the field plate voltage VFP between the field plate FP and the source S increases from 0 V to V_{fp-on} .

(66) FIG. 6 is a time chart showing a control waveform of the semiconductor device 1 according to the first embodiment. A vertical axis represents a voltage, and a horizontal axis represents a time. FIG. 6 shows the control signal V_g , the source-drain voltage V_{ds} , and the inter-terminal voltage VCB of the capacitor CB.

(67) As shown in FIG. 6, the control signal V_g rises at the time T1 and drops at the time T2. When the control signal V_g exceeds a gate threshold voltage of the switching element SD, the switching element SD is turned on, and the source-drain voltage V_{ds} decreases from the off-voltage V_{d-off} to, for example, the on-voltage V_{d-on} close to 0 V. In response to this, the inter-terminal voltage VCB of the capacitor CB decreases to a level of V_{c-on} ($=V_{fp-on}$). At this time, the voltage V_{fp-on} applied between the field plate FP and the source S is favorably higher than the “High” level of the control signal V_g . In order to apply a sufficient level of voltage V_{fp-on} between the field plate FP and the source S, for example, a capacitance value of the capacitor CB is favorably larger than the parasitic capacitance between the field plate FP and the drain D.

(68) When the control signal V_g decreases to a level lower than the gate threshold voltage of the switching element SD, the switching element SD is turned off, and the source-drain voltage V_{ds} increases from the on-voltage V_{d-on} to the off-voltage V_{d-off} . Accordingly, the charging of the capacitor CB is started, and the inter-terminal voltage VCB increases to V_{c-off} .

(69) FIG. 7 is a circuit diagram showing the control circuit CC according to the first embodiment. The control circuit CC includes, for example, bias terminals TB1 and TB2, an input-side terminal TI, and output-side terminals TO1 and TO2. The terminals are separated from one another.

(70) The bias terminal TB1 is connected to, for example, the first terminal DT, and the bias terminal TB2 is connected to, for example, the second terminal ST. The input-side terminal TI corresponds to the control terminal GT. The output-side terminal TO1 is connected to the drain of the first transistor Tr1 and the drain of the second transistor Tr2. The output-side terminal TO2 is connected to the input-side terminal TI, and a signal input to the input-side terminal TI is directly

output from the output-side terminal TO2.

(71) The input-side terminal TI is also connected to the gate of the third transistor Tr3, and the third transistor Tr3 is controlled to be turned on and off by the signal input to the input-side terminal TI.

(72) When the switching element SD to be driven is in the off-state, the first transistor Tr1 of the control circuit CC is in the off-state, and the second transistor Tr2 is in the on-state. Therefore, a voltage substantially equal to the voltage between the drain and the source of the switching element SD is applied between the drain and the source of the first transistor Tr1. Therefore, a breakdown voltage between the drain and the source of the first transistor Tr1 is favorably not less than a breakdown voltage between the drain and the source of the switching element SD.

(73) When the switching element SD is in the off-state, the third transistor Tr3 is in the off-state. Therefore, a voltage substantially equal to the voltage between the drain and the source of the switching element SD is applied between the drain and the source of the third transistor Tr3. Therefore, a breakdown voltage between the drain and the source of the third transistor Tr3 is favorably not less than the breakdown voltage between the drain and the source of the switching element SD.

(74) An input capacitance Ciss, an output capacitance Coss, and a feedback capacitance Crss of the first to third transistors Tr1, Tr2, and Tr3 are favorably smaller than the parasitic capacitances of the switching element SD in order to reduce gate drive loss and switching loss. That is, it is desired that the gate drive loss and the switching loss in the control circuit CC are negligible in comparison with the switching loss of the switching element SD.

(75) FIGS. 8A and 8B are schematic cross-sectional views showing a control method for the semiconductor device 1 according to a comparative example. FIGS. 8A and 8B show a control method for the third electrodes 50 of the switching element SD.

(76) In the example shown in FIG. 8A, the third electrodes 50 are electrically connected to the second electrode 30 in a manner of having a potential same as that of the second electrode 30. In other words, the field plate FP is connected to the source S.

(77) In the example shown in FIG. 8B, the third electrodes 50 are connected to the control terminal GT. That is, the field plate FP has a potential same as that of the gate G. For example, when a “High” level control signal V_g is applied from the control terminal GT to the control electrodes 40 and the switching element SD is turned on, an n-type accumulation layer AL is induced in the first semiconductor layer 11 facing the third electrode 50 (the field plate FP). Accordingly, an electrical resistance of the first semiconductor layer 11 between the adjacent trenches TG is reduced, and the on-resistance of the switching element SD can be reduced.

(78) FIGS. 9A to 9C are graphs showing switching characteristics of the semiconductor device 1 according to the first embodiment and the comparative example. FIGS. 9A to 9C show voltage waveforms and current waveforms at the time of turning on the semiconductor device 1. A horizontal axis represents a time.

(79) FIG. 9A shows turn-on characteristics when the field plate FP is connected to the source S. At the time T1, when the control signal V_g rises and exceeds the gate threshold voltage of the switching element SD, the drain current I_d starts to flow and increases to a level of an on-current I_{d-on} . On the other hand, the source-drain voltage V_{ds} decreases from the off-voltage V_{d-off} to the on-voltage V_{d-on} . During the period, the voltage V_{FP} between the field plate FP and the source S is 0 V.

(80) FIG. 9B shows turn-on characteristics when the field plate FP is connected to the gate G. At the time T1, when the control signal V_g rises and exceeds the gate threshold voltage of the switching element SD, the drain current I_d starts to flow. Since the control signal V_g is also supplied to the field plate FP, the n-type accumulation layer AL is induced in the first semiconductor layer 11 facing the field plate FP, and the on-resistance can be reduced. However, the parasitic capacitance between the gate and the drain is increased, and the rise of the control signal V_g is delayed. As a result, a time for the drain current I_d to reach the level of I_{d-on} and a

time for the source-drain voltage V_{ds} to decrease to V_{d-off} are delayed, and a turn-on time ΔT_{on} is lengthened. That is, when the field plate FP is connected to the gate G, the on-resistance can be reduced, whereas the turn-on time ΔT_{on} becomes longer.

(81) FIG. 9C shows turn-on characteristics in the control method according to the embodiment. At the time T1, when the control signal V_g rises and exceeds the gate threshold voltage of the switching element SD, the drain current I_d starts to flow. On the other hand, the source-drain voltage V_{ds} decreases from the off-voltage V_{d-off} to the on-voltage V_{d-on} . During the period, the voltage V_{FP} between the field plate FP and the source S increases from 0 V to V_{fp-on} . V_{fp-on} is higher than the control signal V_g , and the field plate FP is biased to a potential higher than that of the control signal V_g . Accordingly, the n-type accumulation layer having a higher density is induced in the first semiconductor layer 11, and the on-resistance can be further lowered.

(82) In this case, the inter-terminal voltage VCB of the capacitor CB is applied to the field plate FP, and the capacitance between the field plate FP and the drain is disconnected from the capacitance between the gate and the drain. Therefore, the parasitic capacitance between the gate and the drain can be prevented from increasing. Accordingly, the turn-on time ΔT_{on} can be shortened as compared with the case where the field plate FP is connected to the gate G. The turn-on time ΔT_{on} in the control method according to the embodiment is equal to the turn-on time ΔT_{on} when the field plate FP is connected to the source S.

(83) FIGS. 10A to 10C are graphs showing another switching characteristic of the semiconductor device 1 according to the first embodiment and the comparative example. FIGS. 10A to 10B show voltage waveforms and current waveforms at the time of turning off the semiconductor device 1. A horizontal axis represents a time.

(84) FIG. 10A shows turn-off characteristics when the field plate FP is connected to the source S. At the time T2, when the control signal V_g drops and approaches the gate threshold voltage of the switching element SD, the drain current I_d starts to decrease and decreases to 0 level. On the other hand, the source-drain voltage V_{ds} increases from the on-voltage V_{d-on} to the off-voltage V_{d-off} . During the period, the voltage V_{FP} between the field plate FP and the source S is 0 V.

(85) FIG. 10B shows turn-off characteristics when the field plate FP is connected to the gate G. At the time T2, when the control signal V_g drops and approaches the gate threshold voltage of the switching element SD, the drain current I_d starts to decrease and decreases to the 0 level. The n-type accumulation layer AL induced in the first semiconductor layer 11 disappears as the control signal V_g decreases. In this case, since the parasitic capacitance between the gate and the drain is large, the drop of the control signal V_g is also delayed. As a result, a time for the drain current I_d to reach the 0 level from I_{d-on} and a time for the source-drain voltage V_{ds} to rise from V_{d-on} to V_{d-off} are delayed, and a turn-off time ΔT_{off} is lengthened.

(86) FIG. 10C shows turn-off characteristics in the control method according to the embodiment. At the time T2, when the control signal V_g drops and approaches the gate threshold voltage of the switching element SD, the drain current I_d starts to decrease. The source-drain voltage V_{ds} also increases from the on-voltage V_{d-on} to the off-voltage V_{d-off} . During the period, the voltage V_{FP} decreases from V_{fp-on} to 0 V. Therefore, the n-type accumulation layer induced in the first semiconductor layer 11 disappears.

(87) In the control method according to the embodiment, when the switching element SD is turned off, the electrical connection between the field plate FP and the capacitor CB is cut off, and the field plate FP is connected to the source S. Therefore, the charges are discharged from the field plate FP to the source S, and the field plate voltage V_{FP} decreases from V_{fp-on} to 0 V. Therefore, the discharge of the charges from the field plate FP does not affect the turn-off of the switching element SD, and the turn-off time ΔT_{off} can be shortened.

(88) Therefore, in the semiconductor device 1, a so-called boot strap circuit including the capacitor CB is provided in the control circuit CC, and the voltage V_{FP} between the field plate FP and the source S is controlled. The capacitor CB is charged from the drain side when the switching element

SD is turned off, and when the switching element SD is turned on, the charges are charged to the field plate FP. That is, since the potential of the field plate FP is controlled independently of the gate electrode, the turn-on time and the turn-off time can be shortened, and the loss of a gate driver can be avoided. Further, by biasing the field plate FP to a potential higher than the potential of the gate electrode, the density of the n-type accumulation layer AL can be increased to further reduce the on-resistance.

(89) FIG. 11 is a circuit diagram showing a semiconductor device 2 according to a variation of the first embodiment. The semiconductor device 2 includes, for example, a switching element SD and a control circuit CC2.

(90) In the example, the second diode D2 (see FIG. 2) between the first terminal DT and the third transistor Tr3 is replaced with a resistor R1 in the control circuit CC2. That is, when a switching speed of the first to third transistors Tr1, Tr2, and Tr3 in the control circuit CC2 is sufficiently higher than a switching speed of the switching element SD, the circuit can be simplified and reduced in cost using the resistor R1 instead of the second diode D2.

(91) FIG. 12 is a circuit diagram showing a control circuit CC3 according to another variation of the first embodiment. In the example, delay circuits DE are separately provided between the gate of the first transistor Tr1 and the drain of the third transistor Tr3 and between the gate of the second transistor Tr2 and the drain of the third transistor Tr3.

(92) FIG. 13 is a time chart showing a control method for the semiconductor device 1 according to another variation of the first embodiment. FIG. 13 shows the control signal V_g , the source-drain voltage V_{ds} , the drain current I_d , and the voltage V_{FP} between the field plate FP and the source S.

(93) As shown in FIG. 13, the control signal V_g input to the control terminal GT increases, for example, from Low to High at the time T1. The control signal V_g is applied between the gate G and the source S of the switching element SD (see FIG. 2). Accordingly, the switching element SD is turned on, and the drain current I_d increases from the 0 level to the on-current I_{d-on} .

Accordingly, the source-drain voltage V_{ds} decreases from the off-voltage V_{d-off} to the on-voltage V_{d-on} .

(94) Further, at the time T2 after the time T1, the control signal V_g is decreased from High to Low. Accordingly, the switching element SD is turned off, and the drain current I_d decreases from the on-current I_{d-on} to the 0 level. Accordingly, the source-drain voltage V_{ds} increases from the on-voltage V_{d-on} to the off-voltage V_{d-off} .

(95) On the other hand, the voltage V_{FP} between the field plate FP and the source S is controlled to increase from the 0 level to V_{fp-on} at a time T3 after the time T1 and before the time T2. Such rising control is performed in the control circuit CC3. That is, delay between the time T1 and the time T3 is controlled by the delay circuits DE.

(96) For example, when the first transistor Tr1 is turned on due to noise or the like of the control signal V_g while the source-drain voltage remains at the off-voltage V_{d-off} , V_{fp-on} is applied to the field plate FP. Therefore, dielectric breakdown may occur between the field plate FP and the gate G. Therefore, it is favorable to increase the field plate voltage V_{FP} to V_{fp-on} after a timing at which the switching element SD is turned on. A Schmitt trigger circuit, for example, can be used as the delay circuit DE for delaying the rise of the field plate voltage V_{FP} .

Second Embodiment

(97) FIG. 14 is a schematic cross-sectional view showing a semiconductor device 3 according to a second embodiment. The semiconductor device 3 has a structure in which the switching element SD and the capacitor CB are monolithically integrated. The switching element SD has a structure same as that of the MOS transistor shown in FIG. 1.

(98) As shown in FIG. 14, the semiconductor portion 10 includes a switching region SDR and a capacitor region CBR between the first electrode 20 and the second electrode 30. The switching region SDR and the capacitor region CBR are arranged in a direction along a back surface of the semiconductor portion 10. The second electrode 30 is provided on a surface of the semiconductor

portion **10** via, for example, interlayer insulating films **45** in the capacitor region CBR.

(99) A first trench TG1 is provided in the switching region SDR, and a second trench TG2 is provided in the capacitor region CBR. The first trench TG1 includes the control electrodes **40** and the third electrodes **50**. The second trench TG2 includes fourth electrodes **60**. The fourth electrodes **60** face the semiconductor portion **10** via dielectric films **63**. The fourth electrode **60** is electrically insulated from the semiconductor portion **10** by the dielectric film **63**. The dielectric film **63** is an insulating film having a predetermined dielectric constant.

(100) In the capacitor region CBR, the semiconductor portion **10** includes, for example, the first semiconductor layer **11**, the fifth semiconductor layer **19**, and a second conductivity type sixth semiconductor layer **21**. The first semiconductor layer **11** extends between the first electrode **20** and the second electrode **30**. The fifth semiconductor layer **19** is provided between the first electrode **20** and the first semiconductor layer **11**. The sixth semiconductor layer **21** is provided in the first semiconductor layer **11**. The first semiconductor layer **11** extends between the sixth semiconductor layer **21** and the second electrode **30**. A part of the first semiconductor layer **11** is interposed between the fifth semiconductor layer **19** and the sixth semiconductor layer **21**.

(101) The second trench TG2 extends from a surface of the semiconductor portion **10** on a second electrode **30** side into the first semiconductor layer **11**. A bottom of the second trench TG2 is located in the sixth semiconductor layer **21**. The fourth electrode **60** is provided between the first electrode **20** and the second electrode **30**, and faces the sixth semiconductor layer **21** via the dielectric film **63**. The fourth electrode **60** is connected to the second electrode **30** on an opening side of the second trench TG2. The second electrode **30** is electrically connected to the fourth electrode through a contact hole formed in the interlayer insulating film **45**.

(102) The sixth semiconductor layer **21** is electrically connected to, for example, the first electrode **20**. In other words, the sixth semiconductor layer **21** is electrically connected to the first electrode **20** without interposing a P-N junction.

(103) In the example, the capacitor CB is provided, for example, between the first semiconductor layer **11** and the fourth electrode **60**. The dielectric film **63** is provided such that the capacitor CB has a predetermined capacitance value. Furthermore, the first semiconductor layer **11** and the sixth semiconductor layer **21** constitute the first diode D1 (see FIG. 2). The sixth semiconductor layer **21** is an anode of the first diode D1 and is electrically connected to the first electrode **20**. The first semiconductor layer **11** also serves as a cathode of the first diode D1 and the terminal CBD of the capacitor CB (see FIG. 2).

(104) The first electrode **20** is a drain electrode of the switching element SD and is connected to the anode of the first diode D1 (see FIG. 2). The second electrode **30** is a source electrode of the switching element SD and also serves as the terminal CBS of the capacitor CB (see FIG. 2).

(105) The semiconductor device **3** further includes a diode terminal TD1. The diode terminal TD1 is a cathode terminal of the first diode D1 and is electrically connected to the first semiconductor layer **11**. The diode terminal TD1 is connected to, for example, the source of the first transistor Tr1 (see FIG. 2).

(106) FIGS. 15A and 15B are schematic cross-sectional views showing electrode connections in the semiconductor device **3** according to the second embodiment. FIG. 15A illustrates a connection structure between the first electrode **20** and the sixth semiconductor layer **21**. FIG. 15B illustrates a connection structure between the diode terminal TD1 and the first semiconductor layer **11**.

(107) As shown in FIG. 15A, the first electrode **20** has a contact portion **20p** extending inside a contact trench BT provided on a back surface side of the semiconductor portion **10**. The contact trench BT has a depth reaching the sixth semiconductor layer **21** through the fifth semiconductor layer **19** and the first semiconductor layer **11**. The first electrode **20** is connected to the sixth semiconductor layer **21** at the contact portion **20p** extending into the contact trench BT.

(108) As shown in FIG. 15B, the diode terminal TD1 is provided on the first semiconductor layer **11** via the second insulating film **45**. The diode terminal TD1 is provided on the second insulating

film **45** in a manner of being separated from the second electrode **30**. The diode terminal TD1 has a portion extending into a contact hole **45ch** formed in the second insulating film **45**. That is, the diode terminal TD1 is connected to the first semiconductor layer **11** through the contact hole **45ch**.

(109) FIG. **16** is a schematic cross-sectional view showing a semiconductor device **4** according to a variation of the second embodiment. In the example, the semiconductor device **4** also has a structure in which the switching element SD and the capacitor CB are monolithically integrated. The switching element SD has a structure same as that of the MOS transistor shown in FIG. **1**.

(110) The semiconductor device **4** includes the first electrode **20**, the second electrode **30**, the control electrode **40**, the third electrode **50**, the fourth electrode **60** and a fifth electrode **70**. The semiconductor portion **10** is located between the first electrode **20** and the second electrode **30** and between the first electrode **20** and the fifth electrode **70**, and includes a switching region SDR and a capacitor region CBR. The fifth electrode **70** is provided on the interlayer insulating film **45** on a surface side of the capacitor region CBR. The second electrode **30** and the fifth electrode **70** are provided on the interlayer insulating film **45** in a manner of being separated from each other.

(111) The control electrodes **40** and the third electrodes **50** are provided inside the first trench TG1 in the switching region SDR. The fourth electrode **60** is provided inside the second trench TG2 in the capacitor region CBR. The fourth electrode **60** is connected to the second electrode **30** through, for example, a contact hole formed in the interlayer insulating film **45**.

(112) In the capacitor region CBR, the semiconductor portion **10** includes the first semiconductor layer **11**, the fifth semiconductor layer **19**, the sixth semiconductor layer **21**, a second conductivity type seventh semiconductor layer **22**, and a first conductivity type eighth semiconductor layer **23**. The fifth semiconductor layer **19** is provided between the first semiconductor layer **11** and the first electrode **20**.

(113) The sixth semiconductor layer **21** is provided in the first semiconductor layer **11** between the first electrode **20** and the second electrode **30** and between the first electrode **20** and the fifth electrode **70**. The first semiconductor layer **11** extends between the sixth semiconductor layer **21** and the second electrode **30**. A part of the first semiconductor layer **11** is interposed between the fifth semiconductor layer **19** and the sixth semiconductor layer **21**.

(114) The sixth semiconductor layer **21** is led out to a surface side of the semiconductor portion **10** between the first electrode **20** and the fifth electrode **70**. On the surface side of the semiconductor portion **10**, the sixth semiconductor layer **21** is electrically connected to the fifth electrode **70** via the seventh semiconductor layer **22**. The seventh semiconductor layer **22** is provided between the sixth semiconductor layer **21** and the fifth electrode **70** and contains a second conductivity type impurity having a concentration higher than a concentration of a second conductivity type impurity of the sixth semiconductor layer **21**. The fifth electrode **70** is connected to the seventh semiconductor layer **22** through a contact hole formed in the interlayer insulating film **45**.

(115) The eighth semiconductor layer **23** is provided between the first semiconductor layer **11** and the fifth electrode **70**. For example, the eighth semiconductor layer **23** is provided in the first semiconductor layer **11** on the surface side of the semiconductor portion **10**. The eighth semiconductor layer **23** contains a first conductivity type impurity having a concentration higher than a concentration of a first conductivity type impurity in the first semiconductor layer **11**. The fifth electrode **70** is connected to the eighth semiconductor layer **23** through another contact hole formed in the interlayer insulating film **45**. The fifth electrode **70** is electrically connected to the first semiconductor layer **11** via the eighth semiconductor layer **23**.

(116) In the example, the first electrode **20** is also a drain electrode of the switching element SD. The second electrode **30** is a source electrode of the switching element SD and also serves as the source-side terminal CBS of the capacitor CB.

(117) The first semiconductor layer **11** and the sixth semiconductor layer **21** constitute the first diode D1 (see FIG. **2**). The fifth electrode **70** is an anode electrode of the first diode D1, and is electrically connected to the first electrode **20** via the eighth semiconductor layer **23**, the first

semiconductor layer **11**, and the fifth semiconductor layer **19**.

(118) The capacitor CB is provided between the first semiconductor layer **11** and the fourth electrode **60**. The first semiconductor layer **11** also serves as a cathode of the first diode D1 and the terminal CBD of the capacitor CB (see FIG. 2). The diode terminal TD1 is electrically connected to the first semiconductor layer **11**. The diode terminal TD1 is a cathode terminal of the first diode D1, and is connected to, for example, the source of the first transistor Tr1.

Third Embodiment

(119) FIG. 17 is a schematic view showing a semiconductor device 5 according to a third embodiment. The semiconductor device 5 includes, for example, a switching element SD2 and the control circuit CC. The switching element SD2 is, for example, a power MOS transistor having a planar gate structure. The semiconductor device 5 is, for example, a hybrid device including the switching element SD2 and a control chip including the control circuit CC.

(120) The switching element SD2 includes, for example, the semiconductor portion **10**, a third electrode (hereinafter, referred to as the gate electrode **40**), and a fourth electrode (hereinafter, referred to as the field plate **50**). In the following description, a source electrode is the source S, a drain electrode is the drain D, a gate terminal is the gate G, and a field plate terminal is an FP terminal.

(121) As shown in FIG. 17, the semiconductor portion **10** includes the n-type drift layer **11**, the p-type body layer **13**, the n-type source layer **15**, the p-type contact layer **17**, the n-type buffer layer **19**, a p-type well **25**, and an n-type contact layer **26**.

(122) The p-type well **25** contains a p-type impurity having a concentration lower than a concentration of a p-type impurity of the p-type body layer **13**. The n-type drift layer **11** and the p-type body layer **13** are arranged on the p-type well **25**. The n-type drift layer **11** and the p-type body layer **13** are arranged in a direction along the surface of the semiconductor portion **10**, for example, in the X-direction. The n-type buffer layer **19** is partially provided on the n-type drift layer **11**. The n-type contact layer **26** is provided on the n-type buffer layer **19**. The n-type contact layer **26** contains an n-type impurity having a concentration higher than a concentration of an n-type impurity in the n-type buffer layer **19**.

(123) The n-type source layer **15** and the p-type contact layer **17** are arranged, for example, in the X-direction on the p-type body layer **13**. The n-type source layer **15** is located, for example, between the n-type drift layer **11** and the p-type contact layer **17**. The p-type body layer **13** extends between the n-type drift layer **11** and the n-type source layer **15**.

(124) A thickness of the n-type drift layer **11** in the Z-direction is, for example, larger than a thickness of the p-type body layer **13** in the Z-direction. Shallow trench isolation (STI) is partially provided on the n-type drift layer **11** on the surface side of the semiconductor portion **10**. The STI is provided between a portion of the n-type drift layer **11** facing the p-type body layer **13** and the n-type contact layer **26**.

(125) The gate electrode **40** is provided on the surface of the semiconductor portion **10** via the gate insulating film **43**. The gate electrode **40** faces the p-type body layer **13** between the n-type drift layer **11** and the n-type source layer **15** via the gate insulating film **43**. The field plate **50** is provided on the surface of the semiconductor portion **10** via the FP insulating film **53**. The field plate **50** faces the n-type drift layer **11** via the FP insulating film **53**. The gate electrode **40** and the field plate **50** are provided adjacent to each other, and are separated from each other. The field plate **50** extends on the STI.

(126) The switching element SD2 further includes the interlayer insulating film **45**. The interlayer insulating film **45** is provided on the surface side of the semiconductor portion **10** and covers the gate electrode **40** and the field plate **50**. The source S, the drain D, the gate G, and the FP terminal are provided on the interlayer insulating film **45**.

(127) The drain D is connected to the n-type contact layer **26** through a contact hole formed in the interlayer insulating film **45**. The source S is connected to the n-type source layer **15** and the p-type

contact layer **17** through another contact hole formed in the interlayer insulating film **45**. The gate G and the FP terminal are respectively connected to the gate electrode **40** and the field plate **50** through another contact hole formed in the interlayer insulating film **45**.

(128) The drain D is connected to the first terminal DT. The source S is connected to the second terminal ST. The gate G is connected to the control terminal GT.

(129) The control circuit CC includes the capacitor CB, the first transistor Tr**1**, the second transistor Tr**2**, and the third transistor Tr**3**, and has a configuration same as that of the control circuit CC shown in FIG. **2**. The FP terminal of the switching element SD**2** is connected to the drain of the first transistor Tr**1** and the drain of the second transistor Tr**2**. The control terminal GT is also connected to the gate of the third transistor Tr**3**.

Fourth Embodiment

(130) FIG. **18** is a schematic view showing a semiconductor device **6** according to a fourth embodiment. The semiconductor device **6** includes the switching element SD**2** (see FIG. **17**), the first transistor Tr**1**, the second transistor Tr**2**, the third transistor Tr**3**, the first diode D**1**, the second diode D**2**, and the capacitor CB, which are monolithically integrated. Each element has, for example, a silicon on insulator (SOI) structure provided on a semiconductor substrate SS via an insulating layer **65**. The first transistor Tr**1** to the third transistor Tr**3** are, for example, MOS transistors having a planar gate structure.

(131) The drain D of the switching element SD**2** is connected to the first terminal DT. An anode A of the first diode D**1** and the anode of the second diode D**2** are also connected to the first terminal DT.

(132) The sources S of the switching element SD**2**, the second transistor Tr**2**, and the third transistor Tr**3** are connected to the second terminal ST. One terminal CBS (see FIG. **2**) of the capacitor CB is also connected to the second terminal ST.

(133) The source S of the first transistor Tr**1** is connected to the cathode of the first diode D**1** and the other terminal CBD of the capacitor CB (see FIG. **2**). The drain D of the first transistor Tr**1** is connected to the drain D of the second transistor Tr**2** and the FP terminal of the switching element SD**2**.

(134) The gate G of the switching element SD**2** is connected to the control terminal GT. The gate G of the first transistor Tr**1** and the gate G of the second transistor Tr**2** are connected to the drain D of the third transistor Tr**3** and a cathode K of the second diode D**2**. The gate G of the third transistor Tr**3** is connected to the control terminal GT.

(135) In the semiconductor devices **5** and **6** according to the embodiments, the switching element SD**2** is also operated by the control method shown in FIG. **2**, and the on-resistance and the switching loss can also be reduced.

(136) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions. Additionally, the embodiments described above can be combined mutually.

(137) Embodiments include the following aspects.

Note 1

(138) A control circuit comprising: a first bias terminal; a second bias terminal separated from the first bias terminal; an input-side terminal separated from the first bias terminal and the second bias terminal; a diode having an anode connected to the first bias terminal; a capacitor including a first terminal connected to a cathode of the diode and a second terminal electrically connected to the second bias terminal; a first transistor including a third terminal connected to the first terminal of the capacitor, a fourth terminal, and a first control terminal configured to turn on and off electrical

conduction between the third terminal and the fourth terminal; a second transistor including a fifth terminal connected to the fourth terminal of the first transistor, a sixth terminal connected to the second bias terminal, and a second control terminal configured to turn on and off electrical conduction between the fifth terminal and the sixth terminal; and an output-side terminal separated from the first bias terminal, the second bias terminal, and the input-side terminal and connected to the fourth terminal of the first transistor and the fifth terminal of the second transistor, a control signal based on a signal input to the input-side terminal being input to the first control terminal and the second control terminal, and the first transistor and the second transistor being alternately turned on and alternately turned off.

Note 2

(139) The circuit according to note 1, further comprising: a second output-side terminal separated from the first bias terminal, the second bias terminal, the input-side terminal, and the output-side terminal and configured to directly output a signal input to the input-side terminal.

Note 3

(140) The circuit according to claim 1 or 2, wherein the fourth terminal is a drain-side terminal of the first transistor, and the fifth terminal is a drain-side terminal of the second transistor.

Note 4

(141) The circuit according to any one of notes 1 to 3, further comprising: a second diode having an anode connected to the first bias terminal; and a third transistor including a seventh terminal connected to a cathode of the second diode, an eighth terminal connected to the second bias terminal, and a third control terminal configured to turn on and off electrical conduction between the seventh terminal and the eighth terminal, the first control terminal of the first transistor and the second control terminal of the second transistor being connected to the seventh terminal of the third transistor, and the third control terminal of the third transistor being connected to the input-side terminal.

Note 5

(142) The circuit according to any one of notes 1 to 3, further comprising: a third transistor including a seventh terminal connected to the first bias terminal via a resistor, an eighth terminal connected to the second bias terminal, and a third control terminal configured to turn on and off electrical conduction between the seventh terminal and the eighth terminal, the first control terminal of the first transistor and the second control terminal of the second transistor being connected to the seventh terminal of the third transistor, and the third control terminal of the third transistor being connected to the input-side terminal.

Note 6

(143) A semiconductor device comprising: the control circuit according to any one of notes 1 to 5; and a switching element connected to the control circuit, the switching element including a first electrode connected to the first bias terminal of the control circuit, a second electrode connected to the second bias terminal of the control circuit, a third electrode connected to the output-side terminal of the control circuit, a control electrode connected to the input-side terminal of the control circuit, and a semiconductor portion electrically connected to the first electrode and the second electrode, the control electrode being configured to turn on and off electrical conduction in the semiconductor portion between the first electrode and the second electrode according to the signal input to the input-side terminal, and the third electrode being provided between the first electrode and the second electrode, and facing the semiconductor portion via an insulating film.

Note 7

(144) The device according to note 6, wherein the capacitor of the control circuit has a capacitance value larger than a parasitic capacitance between the third electrode and the first electrode.

Note 8

(145) The device according to note 7, wherein the capacitor of the control circuit is integrated on the semiconductor portion.

Note 9

(146) The device according to any one of notes 6 to 8, wherein the control circuit is integrated on the semiconductor portion.

Note 10

(147) The device according to any one of notes 6 to 8, wherein the control circuit is configured to increase a potential of the third electrode after the switching element is in an on-state.

Claims

1. A control circuit comprising: a first bias terminal; a second bias terminal separated from the first bias terminal; an input-side terminal separated from the first bias terminal and the second bias terminal; a diode having an anode connected to the first bias terminal; a capacitor including a first terminal connected to a cathode of the diode and a second terminal electrically connected to the second bias terminal; a first transistor including a third terminal connected to the first terminal of the capacitor, a fourth terminal, and a first control terminal configured to turn on and off electrical conduction between the third terminal and the fourth terminal; a second transistor including a fifth terminal connected to the fourth terminal of the first transistor, a sixth terminal connected to the second bias terminal, and a second control terminal configured to turn on and off electrical conduction between the fifth terminal and the sixth terminal; a first output-side terminal separated from the first bias terminal, the second bias terminal, and the input-side terminal and connected to the fourth terminal of the first transistor and the fifth terminal of the second transistor; and a second output-side terminal, a control signal based on a signal input to the input-side terminal being input to the first control terminal and the second control terminal, and the first transistor and the second transistor being alternately turned on and off, and the second output-side terminal being separated from the first bias terminal, the second bias terminal, the input-side terminal, and the first output-side terminal and directly outputting a signal input to the input-side terminal.
2. The circuit according to claim 1, wherein the fourth terminal is a drain-side terminal of the first transistor, and the fifth terminal is a drain-side terminal of the second transistor.
3. The circuit according to claim 1, further comprising: a second diode having an anode connected to the first bias terminal; and a third transistor including a seventh terminal connected to a cathode of the second diode, an eighth terminal connected to the second bias terminal, and a third control terminal configured to turn on and off electrical conduction between the seventh terminal and the eighth terminal, the first control terminal of the first transistor and the second control terminal of the second transistor being connected to the seventh terminal of the third transistor, and the third control terminal of the third transistor being connected to the input-side terminal.
4. The circuit according to claim 1, further comprising: a third transistor including a seventh terminal connected to the first bias terminal via a resistor, an eighth terminal connected to the second bias terminal, and a third control terminal configured to turn on and off electrical conduction between the seventh terminal and the eighth terminal, the first control terminal of the first transistor and the second control terminal of the second transistor being connected to the seventh terminal of the third transistor, and the third control terminal of the third transistor being connected to the input-side terminal.
5. A semiconductor device comprising: a control circuit; and a switching element connected to the control circuit, the control circuit comprising: a first bias terminal, a second bias terminal separated from the first bias terminal, an input-side terminal separated from the first bias terminal and the second bias terminal, a diode having an anode connected to the first bias terminal, a capacitor including a first terminal connected to a cathode of the diode and a second terminal electrically connected to the second bias terminal, a first transistor including a third terminal connected to the first terminal of the capacitor, a fourth terminal, and a first control terminal configured to turn on and off electrical conduction between the third terminal and the fourth terminal, a second transistor

including a fifth terminal connected to the fourth terminal of the first transistor, a sixth terminal connected to the second bias terminal, and a second control terminal configured to turn on and off electrical conduction between the fifth terminal and the sixth terminal, and an output-side terminal separated from the first bias terminal, the second bias terminal, and the input-side terminal and connected to the fourth terminal of the first transistor and the fifth terminal of the second transistor, a control signal based on a signal input to the input-side terminal being input to the first control terminal and the second control terminal, and the first transistor and the second transistor being alternately turned on and off, the switching element including a first electrode connected to the first bias terminal of the control circuit, a second electrode connected to the second bias terminal of the control circuit, a third electrode connected to the output-side terminal of the control circuit, a control electrode connected to the input side input-side terminal of the control circuit, and a semiconductor portion electrically connected to the first electrode and the second electrode, the control electrode being configured to turn on and off electrical conduction in the semiconductor portion between the first electrode and the second electrode according to the signal input to the input side input-side terminal, and the third electrode being provided between the first electrode and the second electrode, and facing the semiconductor portion via an insulating film.

6. The device according to claim 5, wherein the capacitor of the control circuit has a capacitance value larger than a parasitic capacitance between the third electrode and the first electrode.

7. The device according to claim 6, wherein the capacitor of the control circuit is integrated on the semiconductor portion.

8. The device according to claim 6, wherein the control circuit is integrated on the semiconductor portion.

9. The device according to claim 5, wherein the control circuit is configured to increase a potential of the third electrode after the switching element is in an on-state.
